



Sadia Mehrin Rahi

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Profiles

 [Sadia Mehrin Rahi](#)

 [SadiaMehrinRahi](#)

About me

Motivated Computer Science and Engineering graduate with a strong research background in AI and cybersecurity. Experienced in building deep learning systems, deploying ML applications, and designing user-friendly web interfaces. Eager to contribute to impactful projects through technical expertise, collaborative teamwork, and continuous learning.

Education

Bangladesh University of Business and Technology (BUBT)

July 2022 - June 2026

Artificial Intelligence

BSc. Engg in CSE

3.71

Completed all academic coursework with a strong focus on AI and cybersecurity research.

Skills

AI/ML & Data Science

TensorFlow/Keras, scikit-learn, XGBoost, Pandas, NumPy, SHAP, Streamlit

Web Development

HTML, CSS, Bootstrap, JavaScript, MongoDB, MySQL, React, Node.js, Firebase

UI/UX & Design

Figma, Canva, Draw.io

Tools

Git, GitHub

Experience

Cyber Security Research Hub - BUBT

Jan 2026 - Present

Student Researcher

Rupnagar R/A, Dhaka

 [Cybersecurity_Research_Hub](#)

Conducted cybersecurity research including deep learning-based intrusion detection, malicious URL classification, and healthcare threat analysis. Built and deployed ML models, collaborated with senior researchers, and contributed to academic publications.

Projects

RT-DeepNIDS: A Real-Time Hybrid Network Intrusion Detection System for IT and IoT Environments

Led the real-time deployment and web-based monitoring application (Streamlit + Scapy) for a hybrid NIDS evaluated across CIC-IDS-2017, CIC-IDS-2018, and ToN-IoT-v3 datasets. Achieved up to 99.94% accuracy (XGBoost) and 96.60% zero-day detection accuracy on unseen IoT traffic, with SHAP-based explainability and 0.5-second live dashboard refresh.

July 2025 - June 2026

 [RT-DeepNIDS](#)

Explainable Early Sepsis Prediction in ICU Using Self-Attention LSTM and Focal Loss
Developed a novel Deep Attention-LSTM model to predict ICU sepsis 6 hours before clinical onset using the PhysioNet 2019 dataset. Applied focal loss to handle class imbalance and self-attention for feature importance.

Jan-April 2026

A Hierarchical Multi View Graph Learning Framework for Malicious URL Detection
Built a 2-stage graph learning framework to detect and classify cyber threats, achieving 98.91% binary and 93.00% multiclass accuracy.

Jan-April 2026

3D Smart City Simulation

Developed a 3D smart city simulation using Python and PyOpenGL to model urban infrastructure and visualize city planning concepts.

July-Dec 2025

ShopNex: Full-Stack E-Commerce Platform with Integrated Payment gateway

Built a full-stack e-commerce website using React and Express.js with SSLCommerz payment integration, providing a secure and smooth shopping experience. Designed responsive UI in Figma.

July-Dec 2025

[🔗 GitHub](#)

Pharmacy Data Analytics Dashboard

Developed a centralized platform to manage, analyze, and visualize pharmacy data, helping optimize inventory, patient interactions, and operational decisions.

Jan-May 2025

[🔗 GitHub](#)

MediSync Healthcare Application

Developed a mobile healthcare app using Android Studio and Java to manage patient information and streamline healthcare services.

Jan-May 2025

[🔗 GitHub](#)

Data Transfer System Using Li-Fi for Sound Pollution Controller

Designed a Li-Fi based vehicle communication system using LDR sensors and Arduino to reduce noise pollution.

Jan-May 2025

Publications

A Decade of Healthcare Cyber Threats: Empirical Analysis, Evidence-Based Prioritisation, and AI Threat Model

ML4CS 2026

May 2026

Accepted

A Hierarchical Multi View Graph Learning Framework for Malicious URL Detection

IEEE SPICSCON 2026

May 2026

Accepted

Understanding XSS in Depth: Practical Attacks & Analysis Using DVW

Cybersecurity Research Hub

March 2026

[🔗 Blog](#)

Extra Curricular Activities

IEEE BUBT Student Branch

Former Joint General Secretary

Nov 2025 - July 2026

Elected executive member responsible for supporting student development, organizing technical activities, and contributing to IEEE's mission of advancing technology for humanity.

IEEE BUBT WIE Student Branch Affinity Group

Former General Secretary

Jun 2025 - July 2026

IEEE RAS BUBT Student Branch Chapter

Joint General Secretary

Aug 2025 - July 2026

Volunteering

ICPC Asia Dhaka Regional Contest 2025 - BUBT

Technical Team

December 2025

Provided technical operations support for a prestigious regional programming contest; gained experience in event coordination and technical management.

BUBT Inter-University Collaborative Programming Contest (BIUCPC) - 2025

Reception & Registration Assistant

November 2025

Assisted in logistics, participant coordination, and reception & registration activities for an inter-university programming contest; developed teamwork and event management skills.

AI Olympiad 2025

Reception & Registration Assistant

May 2025

Assisted with participant registration, information desk operations, and attendee coordination.

ICPC Asia Dhaka Regional Contest 2023 - BUBT

Gaming Activities Coordinator

October 2023

[!\[\]\(a25a22d88c5882f4a20f36103df86562_img.jpg\) Certificate](#)

Coordinated participants and managed on-site gaming activities to ensure smooth event flow.

Languages

Bangla

Fluent (written and spoken)

English

Proficient in written and spoken English.

Hindi

I have basic conversational skills in Hindi

References

Available upon request